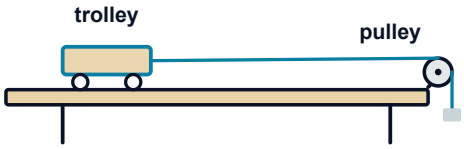
DO NOW · REQUIRED PRACTICAL

Acceleration and force

Forces
Paper 2 · Combined and Separate
6 steps

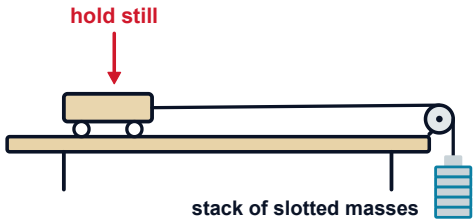
A student investigates how the force on a trolley affects its acceleration.

1 Attach a string to a trolley and pass it over a pulley at the end of the bench.

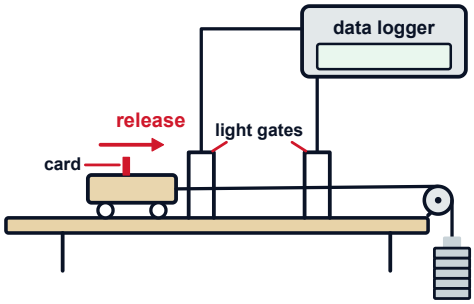


string over the pulley at the end of the bench

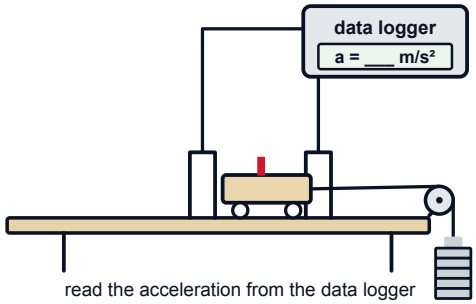
2 Hang a stack of slotted masses on the end of the string and hold the trolley still.



3 Release the trolley so that it passes through two light gates.

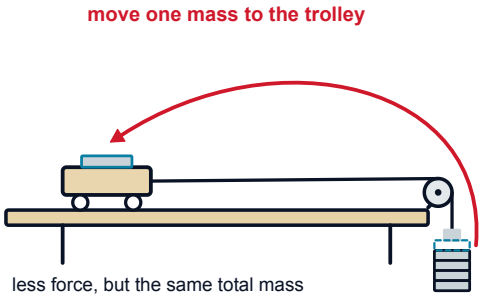


4 Record the acceleration calculated by the data logger.



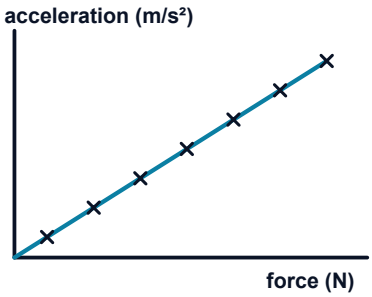
read the acceleration from the data logger

5 Move one slotted mass from the stack to the trolley to reduce the force, then repeat.



less force, but the same total mass

6 Plot a graph of acceleration against force.



force = weight of the hanging masses

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